

AI Agent Evaluation Metrics — Final Summary Table

Source: AI_Agent_Evaluation_Complete (1).pdf. The calculation column follows the formulas/calculation descriptions in the PDF. Important: values described as a "PDF example configuration" are example settings shown in the document, not universal recommended thresholds. The PDF's general rule is: Score $\geq$ Threshold = Pass; Score < Threshold = Fail.

Metric	Good Range	Bad Range	Threshold	How to Calculate	Resource	Definition
Plan Quality (PlanQualityMetric)	$\geq$ threshold; higher is better	< threshold; lower is worse	Not explicitly specified in the PDF. General threshold examples: 0.5 / 0.7	Plan Quality Score = AlignmentScore(Task, Plan) Extract the task and plan from the trace, then use an LLM to generate the alignment score.	DeepEval / supplied evaluation framework	Measures whether the agent's plan is logical, complete, and aligned with the user's task.
Plan Adherence (PlanAdherenceMetric)	$\geq$ threshold; higher is better	< threshold; lower is worse	Not explicitly specified in the PDF	Plan Adherence Score = AlignmentScore((Task, Plan), Execution Steps) Extract the task and plan, compare them with the actual execution steps, then use an LLM to generate the alignment score.	DeepEval / supplied evaluation framework	Measures whether the agent actually follows its own plan during execution.
Tool Correctness (ToolCorrectnessMetric)	≥ 0.5	< 0.5	0.5 default threshold; configurable. Strict mode: threshold becomes 1.	Tool Correctness = Correctly Used Tools / Total Tools Called The score can also consider tool names, input parameters, outputs, ordering, exact matching, and optional LLM-based optimality.	DeepEval	Measures whether the agent selected and used the appropriate/expected tool(s) for the task.
Argument Correctness (ArgumentCorrectnessMetric)	≥ 0.5 by default ≥ 0.7 in the PDF example configuration	< 0.5 by default	0.5 default threshold; the PDF's example configuration uses 0.7. Strict mode: threshold becomes 1.	Argument Correctness = Correctly Generated Input Parameters / Total Number of Tool Calls An LLM determines whether each tool-call argument correctly addresses the input/task.	DeepEval	Measures whether the arguments/input parameters generated for each tool call are correct for the task.
Task Completion (TaskCompletionMetric)	≥ 0.7 in the PDF example configuration	< 0.7 in the PDF example configuration	0.7 in the PDF example configuration	Task Completion Score = AlignmentScore(Task, Outcome) Extract the task and outcome from the trace, then calculate how well the outcome aligns with the task using an LLM-as-a-judge.	DeepEval	Measures whether the agent effectively accomplishes the user's intended task.
Step Efficiency (StepEfficiencyMetric)	≥ 0.7 in the PDF example configuration	< 0.7 in the PDF example configuration	0.7 in the PDF example configuration	Step Efficiency Score = AlignmentScore(Task, Execution Steps) Extract the task and evaluate whether execution steps are necessary and direct. Unnecessary actions reduce the score.	DeepEval	Measures how efficiently the agent executes a task, penalizing duplicate actions, repeated searches, unnecessary retrieval, detours, loops, and speculative actions.